NM2 - cv3

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1 Volterra-Lotkeho systém rovnic

$$u_1(t) = au_1 - bu_1u_2$$
$$u_2(t) = cu_1u_2 - du_2$$

- Interval: $[0, 20]$
- Numerická metoda: Euler
- Integrační krok: 10^{-2}

1.1 Změna v parametru a

Table 1: Analýza v parametru a

a	b	c	d	$u_1(0)$	$u_2(0)$
1.0	1.0	1.0	1.0	1.0	0.5
2.0	1.0	1.0	1.0	1.0	0.5
3.0	1.0	1.0	1.0	1.0	0.5
4.0	1.0	1.0	1.0	1.0	0.5
5.0	1.0	1.0	1.0	1.0	0.5

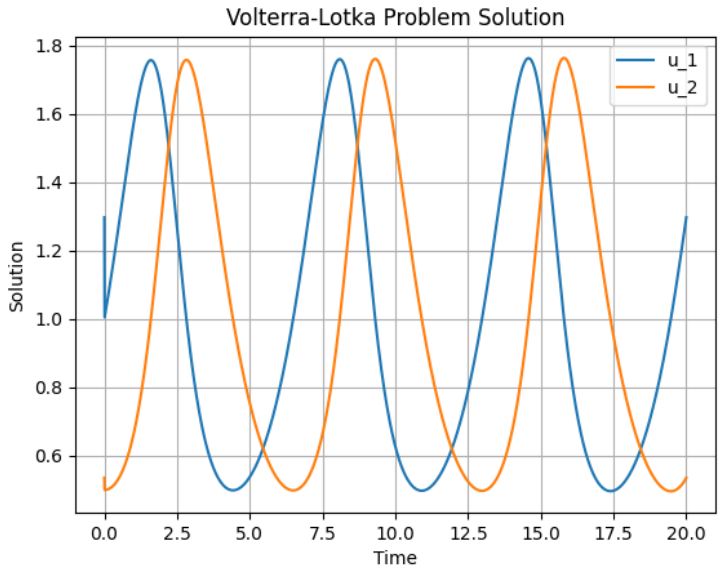

Figure 1: $a = 1$

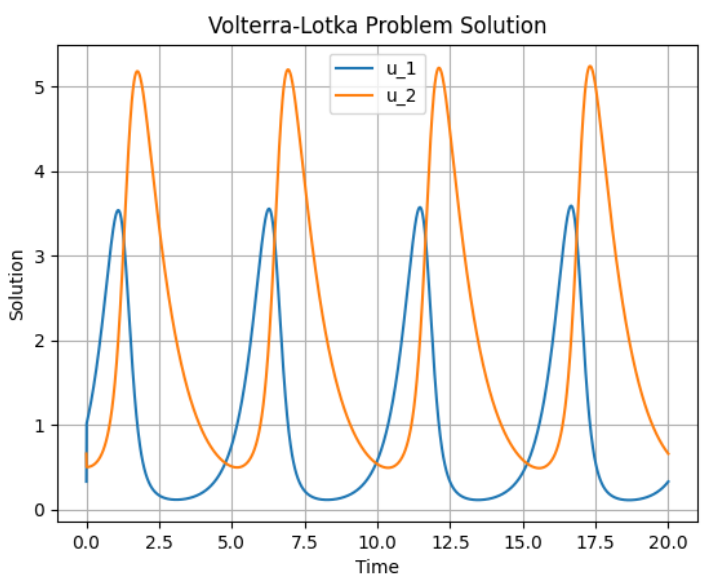

Figure 2: $a = 2$

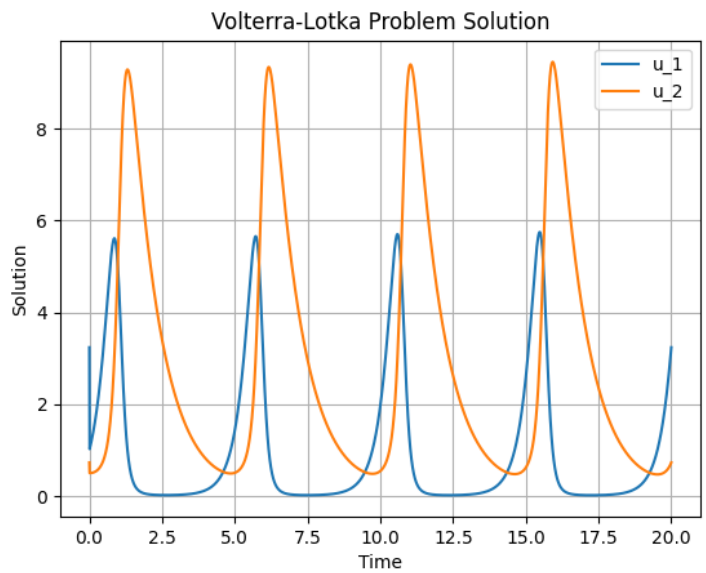

Figure 3: $a = 3$

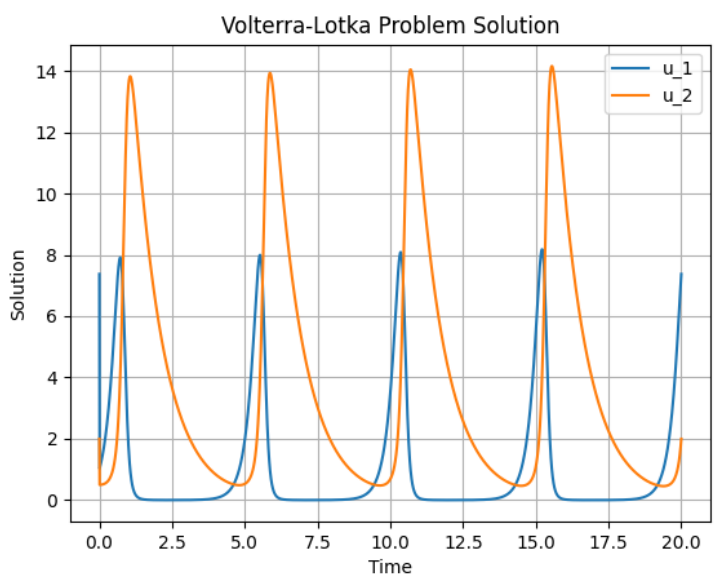

Figure 4: $a = 4$

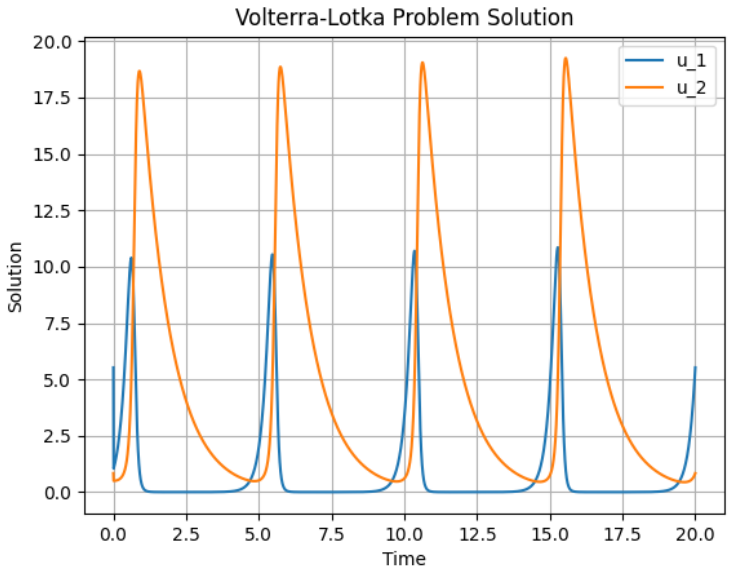
Volterra-Lotka Problem Solution
u_1
u_2
Solution
Time
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

Figure 5: $a = 5$